Mathematics in the Modern World

Prepared by: ENGR. PAJA, M.R.

LOGICAL CONNECTIVES

not

and

or

if.....then

if and only if

Basic Connectives and Truth Tables

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \Rightarrow q$	$p \Leftrightarrow q$
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	0
1	1	1	1	0	1	1

Constructing Truth Tables

- Construct the truth table for the following compound proposition.

$$((p \wedge q) \vee \neg q)$$

p	q	$p \wedge q$	$\neg q$	$((p \wedge q) \vee \neg q)$
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

Precedence of Logical Operators

- As in arithmetic, an ordering is imposed on the use of logical operators in compound propositions.
- However, it is preferable to use parentheses to disambiguate operators and facilitate readability.

$$\neg p \vee q \wedge \neg r \quad \equiv \quad (\neg p) \vee (q \wedge (\neg r))$$

- To avoid unnecessary parenthesis, the following precedences hold:
  1. Negation ($\neg$)
  2. Conjunction ($\wedge$)
  3. Disjunction ($\vee$)
  4. Implication/Conditional ($\rightarrow$)
  5. Biconditional ($\leftrightarrow$)

Usefulness of Logic

- Logic is more precise than natural language.
  - You may have cake or ice cream.
    - Can I have both?
  - If you buy your air ticket in advance, it is cheaper.
    - Are there or not cheap last-minute tickets?

- For this reason, logic is used for hardware and software specification.
  - Given a set of logic statements,
  - One can decide whether or not they are satisfiable (i.e., consistent), although this is a costly process…

s : John studies.
p : John passes.
h : John is happy.

$\wedge$

This symbol replaces the word "**and**".

John passes and John is happy.

p $\wedge$ h

s : John studies.
p : John passes.
h : John is happy.

~

This symbol negates the statement it precedes.

John does not study.

$\sim s$

s : John studies.
p : John passes.
h : John is happy.

$\vee$

This symbol replaces the word "or".

John studies or John does not pass.

$s \vee \sim p$

s : John studies.
p : John passes.
h : John is happy.



This symbol replaces the connective "**if ... then**"

If John studies then John passes.

s → p

s : John studies.
p : John passes.
h : John is happy.

$\leftrightarrow$

This symbol replaces the words "**if and only if**".

John is happy if and only if John studies.

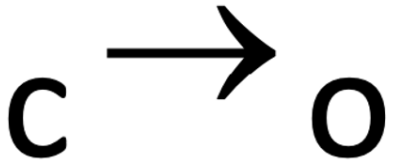
h $\leftrightarrow$ **s**

When using the connective



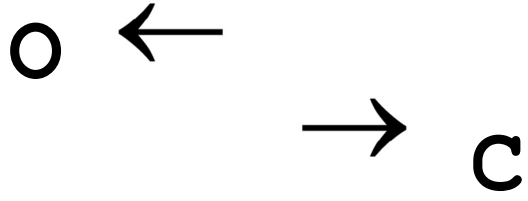
The direction of the arrow is important.

cause  **effect**



If Herman lives in Cleveland then Herman lives in Ohio.

Herman lives in Ohio if Herman lives in Cleveland.



Herman lives in Cleveland only if Herman lives in Ohio.



John is happy only if he studies.



John is happy if he studies.



John is happy if and only if he studies.

and : $\wedge$
or : $\vee$
if...then : $\rightarrow$
if and only if : $\leftrightarrow$
not : **~**

m: Herman takes math.
h: Herman is happy.
a: Herman gets an A.
b: Herman bribes the teacher.

and : $\wedge$
or : $\vee$
if...then : $\rightarrow$
if and only if : $\leftrightarrow$
not : ~

m: Herman takes math.
h: Herman is happy.
a: Herman gets an A.
b: Herman bribes the teacher.

Herman takes math and he is not happy. (m $\wedge$ ~h)

If Herman gets an A then he is happy. (a $\rightarrow$ h)

Herman takes math if and only if he bribes the teacher. (m $\leftrightarrow$ b)

(3 +5) × 2 is different from 3 + (5 × 2)

similarly

l ∧(g → a) is different from (l∧g) → a

- When a statement has two or more connectives, one of the connectives is said to be dominant.
- This determines the grouping (placement of parentheses).

A **SIMPLE STATEMENT** has one verb.
The following four statements are simple:

Today is Friday. **It is hot.**
f **h**

We will swim. **We will picnic.**
s **p**

f **h** **s** **p**

Now we have two **COMPOUND STATEMENTS.**

Today is Friday and it is hot.

We will swim or we will picnic.

f ∧ h s ∨ p

Finally, we can connect these two compound statements:

If Today is Friday and it is hot

then we will swim or we will picnic.

$$[f \wedge h] \rightarrow [s \vee p]$$



is the **dominant connective**. It is last one placed.

and : $\wedge$
or : $\vee$
if...then : $\rightarrow$
if and only if : $\leftrightarrow$
not : ~

l: The teacher likes Herman.
g: Herman does good work.
a: Herman gets an A.

The teacher likes Herman and if Herman does good work then Herman gets an A.

$[\ l\ ] \wedge [\ g \rightarrow a\ ]$

If the teacher likes Herman and Herman does good work then Herman gets an A.

$[\ l \wedge g\ ] \rightarrow [\ a\ ]$

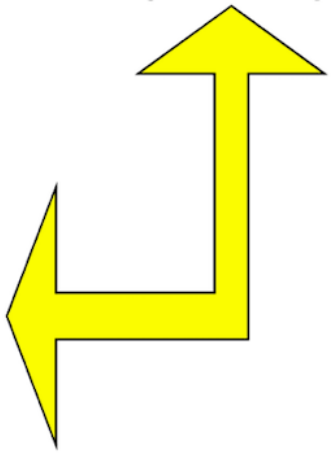
(statement) **DOMINANT CONNECTIVE** (statement)

(3+5) × 2 is different from **3 + (5 × 2)**

Without parentheses
we have a convention
to interpret:

3 + 5 x 2

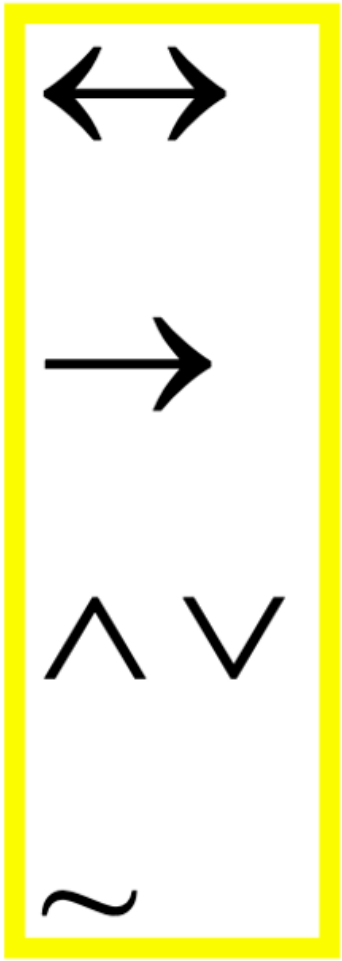
to mean **(3) + (5 x 2)**



When an arithmetic sentence contains **+** and **x** signs and NO PARENTHESES to indicate otherwise we assume that **x** is dominant.

There is a similar convention in logic:

If a statement contains two or more connectives and there are **NO PARENTHESES** to show you how to group symbols, then we will agree that the **DOMINANT CONNECTIVE** is ***whatever is highest on this chart.***



If it rains then we go to a movie and if it doesn't rain then we swim.

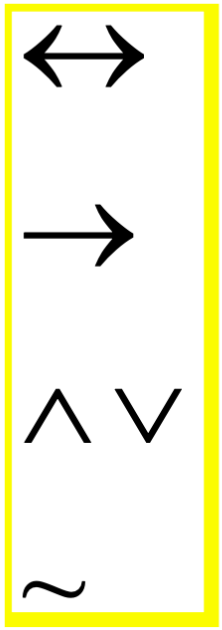
$$(r \rightarrow m) \wedge (\sim r \rightarrow s)$$

r: It rains.

m: We go to a movie.

s: We swim.

In this statement, the parentheses are necessary because "and" is dominant.



If Herman does not study then he will fail math and his life will be ruined.

$$[\sim s] \rightarrow [f \wedge r]$$

s: Herman studies.
f: He will fail math.
r: His life will be ruined.

Because → is the dominant connective, the parentheses are correctly placed…BUT … they are unnecessary.

If Herman does not study then he will fail math and his life will be ruined.

$$\sim s \rightarrow f \wedge r$$

Because this means

$$[\sim s] \rightarrow [f \wedge r]$$

$\leftrightarrow$

$\rightarrow$

$\wedge$ $\vee$

$\sim$

Interpret the following statement:

$$\left(\sim p \rightarrow q \vee r\right) \leftrightarrow \left(\sim s \wedge r\right)$$



This is highest
on the chart.
This is dominant.

p: Pat is lying.
q: Queen is lying.
r: Roy is telling the truth.
s: San is telling the truth.

p: Pat is lying.
q: Queen is lying.
r: Roy is telling the truth.
s: San is telling the truth.

$$(\sim p \rightarrow q \vee r) \leftrightarrow (\sim s \wedge r)$$

If Pat is not lying then Queen is lying or Roy is telling the truth. $(\sim p) \rightarrow (q \vee r)$

Now we have two simpler statements to deal with.

$$(\sim p) \rightarrow (q \vee r)$$

In the pink statement , $\rightarrow$ is highest on the chart.

p: Pat is lying.
q: Queen is lying.
r: Roy is telling the truth.
s: San is telling the truth.

$$[\sim p] \rightarrow [q \vee r] \leftrightarrow [\sim s \wedge r]$$

Now we have two simpler statements to deal with.

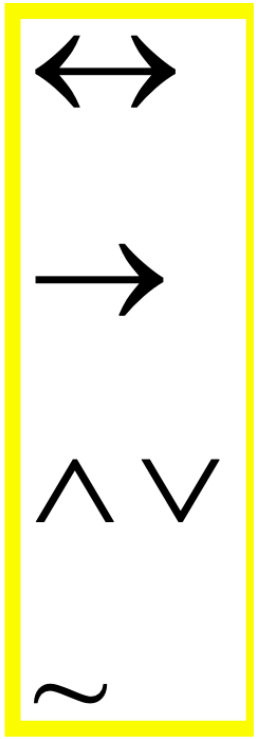
San is not telling the truth and Roy is telling the truth.

$[\sim s] \wedge [r]$

In the blue statement , $\wedge$ is highest on the chart.

p: Pat is lying.
q: Queen is lying.
r: Roy is telling the truth.
s: San is telling the truth.

$$\Big[\big[\sim p\big] \rightarrow \big[q \vee r\big]\Big] \leftrightarrow \Big[\big[\sim s\big] \wedge \big[r\big]\Big]$$



If Pat is not lying then Queen is lying or Roy is telling the truth if and only if San is not telling the truth and Roy is telling the truth.

- A **proposition**, which is a declarative statement with a distinct logical value, can either be simple (atomic) or compound (complex or molecular) propositions.

Type	Logical Operator	Symbolic Form	Read as
Negation	Not	¬p Or ~p	not p
Conjunction	And	p ∧ q	p and q
Disjunction	Or	p ∨ q	p or q
Conditional	If…then	p →	if then q
Bi-conditional	if and only if	p ↔	p if and only if q

Example: Convert the following English sentences to mathematical symbols based on the following simple propositions.

p - Daniel attends the opera concert.
g - Kim attends the opera concert.
r – Lyka performs in the opera concert.
s - Gwen performs in the opera concert.

1. **Either** Daniel **and** Kim attend the opera concert **or** Lyka performs in the opera concert.
2. Daniel attends the opera concert **and either** Kim attends the opera concert **or** Gwen will not perform in the opera concert.
3. Daniel **and** Kim **will not both** attend the opera concert, **but** Lyka and Gwen **will both not** perform in the opera concert.
4. **Either** Daniel **or** Kim attends the opera concert, **but neither** Lyka **nor** Gwen performs in the opera concert.
5. **If** Kim attends the opera concert, **then** Gwen **will not** perform in the concert.
6. Lyka **will not** perform in the opera concert **if and only if** Gwen will perform in the opera concert; **but** Daniel **will not** attend the opera concert.